Discrete random variables

1 State whether X is discrete in the following:

a X is a random variable for the length of a crocodile.

b X is a random variable for the number of games won by Manchester United in a season.

c X is a random variable for the number of television a person owns.

d X is a random variable for the weight of a hippopotamus to the nearest kilogram.

e X is a random variable for the time taken by a driver to complete one lap of a circuit.

f X is a random variable for the length of a queue at a supermarket checkout.

g X is a random variable for the number of pears on each pear tree in an orchard.

2 Out of 100 students sitting an exam, 10 student numbers were drawn randomly for a survey. 35 out of the 100 students are female. Can the number of females in the 10 chosen be represented by a discrete random variable?

3 A multiple choice test contains 10 questions, each with subparts (a) and (b). The answer to each subpart is awarded a half mark if correct, and zero if incorrect. If a student randomly answers each question, can the number of marks gained on this test be modelled by a discrete random variable?

4 Explain why the following situations cannot be represented by a discrete random variable:

a On a popular TV cooking show, contestants are to randomly choose a blue, red or yellow serviette from a box to split themselves into random cooking teams.

b A random number generator generates a real number between 1 and 7 inclusive.

c The time taken to download a 55-minute episode of Here Come the Habibs during off-peak times is anywhere between 5 and 13 minutes.

d At a local bus stop, the time spent waiting for the next bus during peak hour is up to 10 minutes.

e The time between customers using the express checkout at a supermarket is monitored for efficiency purposes. Hence, the operations manager is interested in the checkout operator's waiting time for the next customer to arrive.

f Paul arrives at the train station each morning between 8:30 am and 8:45 am and records the time he arrives for one week.

g The actual capacity of bottles of orange juice, each advertised as containing 1000 mL, are monitored for quality assurance.

h An ice-cream shop is monitoring the popularity of its various flavours of ice-cream by analysing the flavour preference of randomly chosen customers.

5 A photocopier in a busy school breaks on [redacted]ular basis.

a Can the number of breakdowns occuring in a fortnight be represented by a discrete random variable?

b Can the time between breakdowns be represented by a discrete random variable? Explain your answer.

6 The operations manager at a fast food restaurant is monitoring the number of customers arriving at the drive-through on Sunday. During the 24 hours of monitoring, 360 customers used the drive-through.

a Can the number of cars in a randomly chosen 30-minute time period be modelled by a discrete random variable?

b Can the time elapsed between customers arriving at the drive-through on the Sunday be modelled by a discrete random variable?

7 The weights of babies born in a local hospital in the last month have been recorded.

a A midwife is interested in the probability that the next baby born would weigh more than $2.4\ \mathrm{kg}$. If X represents the weight of the next baby born, is X a discrete random variable? Explain your answer.

b The midwife is also interested in the probability that of the next 5 babies born, what number of babies would weigh more than $2.4\ \mathrm{kg}$. If Y represents the number of babies in the next 5 babies born that weigh more than $2.4\ \mathrm{kg}$, is Y a discrete random variable? Explain your answer.

8 The quality control manager of the installation of a fibre-optic network is monitoring the faults found in the cable being used.

a Can the metres between successive faults in the fibre optic cable being analysed be modelled by a discrete random variable?

b Can the number of faults found in a randomly chosen $100\ \mathrm{m}$ length of the fibre optic cable be modelled by a discrete random variable?

9 Lucy has 10 red tea cups, 10 blue tea cups and 10 green cups in her cupboard. She drinks a lot of tea and these tea cups end up in the dishwasher when she's used them.

a Can the colour of tea cup last placed in the dishwasher be modelled by a discrete random variable?

b Can the number of blue tea cups found in the dishwasher at any moment be modelled by a discrete random variable?

10 A coin is tossed three times and the total number of tails is recorded.

a Can this experiment be represented by a discrete random variable?

b If X represents the number of possible tails in the three coin tosses, list all the possible outcomes of the experiment.

11 A random number generator generates an integer between 1 and 6 inclusive. If D represents the integer generated, list all possible outcomes for D.

12 A student designed a game of chance in which two fair tetrahedral dice (both with faces numbered $1, 2, 3$ and 4) were thrown and then the score, X, was calculated from the sum of the numbers appearing uppermost on the dice. List all the possible outcomes of X.

13 Six cards are drawn from a deck of 52 cards, and the number of kings in the draw is recorded. If X represents the number of kings in the draw, list all the possible outcomes of X.

14 The time between customers using the express checkout at Coles is monitored for efficiency purposes. The next 8 customers arriving at the checkout are observed. The operations manager monitors how many times the employee checkout operator waits more than 3 minutes for the next customer to arrive, in this group of 8.
If Y represents the number of times the checkout operator waited more than 3 until the next customer arrives, for the next 8 customers, list all the possible outcomes of Y.

15 The actual capacity of bottles of orange juice, each advertised as containing 1000 mL, are monitored for quality assurance. The operations manager is monitoring the number of bottles that are under the stated amount. A local deli receives a shipment of these bottles and opens a box containing 9 bottles.
If Y represents the number of bottles that are under the advertised capacity in the box of 9 bottles, list all the possible outcomes.

16 The burner on an old gas stove ignites immediately on operation with a probability of 70%. Someone uses this burner 10 times in one day.
If Y represents the number of times the burner ignites immediately that day, list all the possible outcomes.

17 There is a surge of students falling ill from the flu in the boarding house of a school. The chance that a student from the boarding house visiting the school nurse has to be referred to a doctor for the flu is 2 in 5. On Monday morning, 8 students from the boarding house visit the school nurse.
If Y represents the number of students from the boarding house being referred to the doctor for the flu on this Monday morning, list all the possible outcomes of Y.